www.Padasalai.Net www.TrbTnpsc.com

12

Register No. 02

Half-Yearly Examination - 2025

CHEMISTRY

Time : 3.00 Hrs. Marks : 70

PART - I

Answer all questions. 15 x 1 = 15

1. The metal oxide which cannot be reduced to metal by carbon is
   a) PbO b) FeO c) ZnO d) Al_2O_3
2. The basicity of hypophosphorus acid is
   a) 1 b) 2 c) 3 d) 4
3. The highly stable allotrope of carbon is
   a) graphite b) diamond c) fullerenes d) carbon nanotubes
4. Which of the following ions of salt is white in colour
   a) Cd^{+2} b) Cu^{+2} c) Co^{+3} d) V^{3+}
5. Coordination number of Ni in $[Ni(C_2O_4)_3]^{4-}$ is
   a) 3 b) 4 c) 2 d) 6
6. Packing efficiency of body centred cubic (bcc) in unit cell is
   a) 32% b) 48% c) 23% d) 68%
7. Which of the following is correct for first order reaction.
   a) $t_{1/2} \alpha$ (concentration)^0 b) $t_{1/2} \alpha$ (concentration)$^{-1}$ c) $t_{1/2} \alpha$ (concentration)^1 d) $t_{1/2} \alpha$ (concentration)$^{1/2}$
8. $H_2PO_4^-$ the conjugate base of
   a) PO^{3-}_4 b) A_3PO_4 c) P_2O_5 d) HPO^{2-}_4
9. The emf of standard hydrogen electrode (SHE) is
   a) one(1) b) infinite c) zero (0) d) negative (– 1)
10. The most effective electrolyte for the coagulation of As_2S_3 sol is
    a) NaCℓ b) $Ba(NO_3)_2$ c) $K_3[Fe(CN)_6]$ d) $Aℓ_2(SO_4)_3$
11. Williamson synthesis of preparing dimethyl ether is a/an
    a) SN_2 reactions b) SN_1 reactions c) electrophilic addition d) electrophilic substitution
12. Which one of the following is incorrectly matched?
    a) Tollen's reagent - $AgNO_3 + NH_4OH$ b) Fehling's solution - $CuSO_4$ + Roshelle salt
    c) Benedict's solution - $CuSO_4$ + Sodium cetrate + NaOH d) Bayer's reagent - Conc. HCℓ + Anly.$ZnCℓ_2$
13. $CH_3 - CH_2 - Br \xrightarrow[\Delta]{aq.NaOH} A \xrightarrow[\Delta]{KMnOSO_4/H^+} B \xrightarrow{\Delta} NH_3$

    'D' $\xleftarrow{LiAlH_4}$ C

    'D' is a) bromomethane b) α - bromosodium acetate c) methanamine d) Ethanamine
14. Which one given below is a non-reducing disaccharide?
    a) sucrose b) maltose c) lactose d) galactose
15. Assertion : - Paracetamol and aspirin are both antipyretics.
    Reason : - Both paracetamol and aspirin are controlled and reversible loss of conciousness by affecting central nervous sytem.
    a) If both assertion and reason are true and reason is the correct explanation of assertion.
    b) If both assertion and reason are true but reason is not the correct explanation of assertion.
    c) Assertion is true but reason is false d) both assertion and reason are false

PART - II

Answer any six question. Q.No.24 is compulsory. 6 x 2 = 12

Kindly Send Me Your Study Materials To Us Email ID: padasalai.net@gmail.com

www.Padasalai.Net www.TrbTnpsc.com

16. Explain the following terms with suitable examples (i) Gangue (ii) slag
17. What is inert pair effect?
18. Why d-block elements exhibit variable oxidation states?
19. Distinguish between isotropy and anisotropy in solid.
20. Define ionic product of water. Give its value at room temperature.
21. Write a note on electrophorosis.
22. State Popoff's rule.
23. Give any three differences between DNA and RNA.
24. Write the rat law for the following reactions. i) $3A + 5B_2 \rightarrow 4CD$ ii) $X_2 + Y_2 \rightarrow 2XY$

PART - III

Answer any six questions. Q.No.33 is compulsory. **6 x 3 = 18**

25. i) CO is a reducing agent. Justify with an example. (1½)
    ii) Write a note on Fischer - Tropsch synthesis. (1½)
26. Explain froath floatation method.
27. i) Write a note on chromylchloride test.
    ii) Write the electronic configuration of Ce^{+4} and Co^{+2}.
28. In the complex, $[Pt(NO_2)(H_2O)(NH_3)_2]Br$, Identify the following.
    i) central metal ion/atom ii) coordination number iii) oxidation state of central metal ion
    iv) IUPAC name v) Coordination sphere
29. State Faraday's law of electrolysis.
30. Convert the following : i) Ethane-1, 2-diol to Ethanol ii) Glycerol to Glycerose iii) Phenol to benzene
31. i) Define : Isoelectric point. ii) What is epimers? Give example.
32. Write a note on Vulcanization of rubber.
33. Identify A, B and C in the following sequence of reactions.

$$C_6H_5{}^+N_2Cl^- \xrightarrow[Cu_2O/200^\circ C]{NH_3} \text{'A'} \xrightarrow[Pyridine]{C_6H_5COCl} \text{'B'} \xrightarrow{KMnO_4/\Delta(heat)} \text{'C'}$$

PART - IV

Answer all question. **5 x 5 = 25**

34. a) i) What is cyanide leaching? The extraction of gold by leaching with NaCN involves both oxidation and reduction. Justify giving equation. (2½)
    ii) Give the limitations of Ellingam diagram (1½) **(OR)**
    b) Describe the structure of Diborane.
35. a) i) Compare lanthanides and actinides. **(OR)**
    b) Apply the VB theory for the following compounds (i) $[Fe(CN)_6]^{3-}$ (ii) $[COF_6]^{3-}$
36. a) Explain Schottky and Frenkel defect. **(OR)**
    b) Derive an expression for Ostwalds dilution law.
37. a) i) State Kohlrausch's law. How it is used to calculate molar conductance at infinite dilution of a weak electrolyte.
    ii) Explain about Glavanic cell notation. **(OR)**
    b) Write a note on (i) HVZ reaction (ii) Stephen's reaction (iii) Crossed Cannizzaro reaction.
38. a) Explain about reduction of nitrobenzene in different medium. **(OR)**
    b) (i) Write a note on anti-oxidant.
    (ii) Writea note on Buna-S and Nylon-2, 6.

Kindly Send Me Your Study Materials To Us Email ID: padasalai.net@gmail.com